LOW-RESOURCE ASR WITH PRETRAINED SPEECH SELF-SUPERVISED REPRESENTATIONS

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ABSTRACT

This project studies low-resource automatic speech recognition (ASR) with pretrained speech self-supervised learning (SSL) representations. We implement a LibriSpeech pipeline with transcript normalization, JSONL manifests, character-level CTC training, greedy decoding, WER/CER evaluation, and qualitative error analysis. Using approximately one hour from LibriSpeech train-clean-100, we compare a small log-mel CTC baseline with frozen wav2vec 2.0 and HuBERT encoders followed by a lightweight CTC head. A focused layer ablation shows that an intermediate wav2vec 2.0 layer is substantially stronger than the final layer under the frozen-encoder setting. The best system—HuBERT base with a frozen encoder—reaches 0.662 WER and 0.236 CER on the evaluated test-clean set. These findings confirm that pretrained SSL representations are useful in low-resource ASR, and that the choice of hidden layer can matter as much as the choice of SSL model family.

Index Terms— Low-resource ASR, self-supervised learning, wav2vec 2.0, HuBERT, CTC

1. INTRODUCTION

Low-resource automatic speech recognition (ASR) is difficult because the model must learn an acoustic-to-text mapping from limited paired speech and transcripts. In conventional supervised ASR, thousands of hours of labeled speech are used to train end-to-end models [?]. When only a small amount of labeled data is available—one hour or less—the model lacks sufficient supervision to learn robust acoustic-phonetic mappings, leading to poor generalization.

Speech self-supervised learning (SSL) addresses this limitation by pretraining acoustic encoders on large unlabeled audio corpora, producing representations that capture phonetic and lexical structure without requiring transcripts. Models such as wav2vec 2.0 [?] and HuBERT [?] learn via contrastive or masked-prediction objectives on raw audio, and their frozen hidden states have been shown to benefit downstream tasks including phoneme recognition, keyword spotting, and ASR [?].

This project follows that paradigm and builds an English ASR system centered on pretrained wav2vec 2.0 and HuBERT representations. The study asks two main questions: (1) Do pretrained SSL hidden states help in a one-GPU, low-resource LibriSpeech setting where only one hour of labeled speech is available? (2) Does the choice of SSL model and hidden layer matter when the downstream model is deliberately small and the encoder is kept frozen?

Our contributions are: (i) a reproducible end-to-end low-resource ASR pipeline, (ii) a controlled comparison of wav2vec 2.0 and HuBERT against a non-pretrained log-mel baseline, and (iii) a clean layer ablation demonstrating that intermediate SSL layers can be substantially more informative for CTC decoding than the final layer. All experiments run on a single consumer GPU (RTX 4060, 8 GB VRAM), making the setup accessible and reproducible.

2. METHOD

2.1. System Overview

The overall system architecture is shown in Fig. 1. The pipeline takes raw 16 kHz audio waveforms, passes them through a frozen SSL encoder to extract frame-level hidden states, projects these to character logits via a lightweight CTC head, and applies greedy CTC decoding to produce the transcript. For the log-mel baseline, the SSL encoder is replaced with a small feed-forward network over 80-dimensional log-mel filterbank features.

2.2. Data and Normalization

All experiments use LibriSpeech [?], a public-domain English read-speech corpus. To avoid unreliable long downloads from OpenSLR, the data preparation script reads the Hugging Face Parquet mirror (openslr/librispeech_asr),

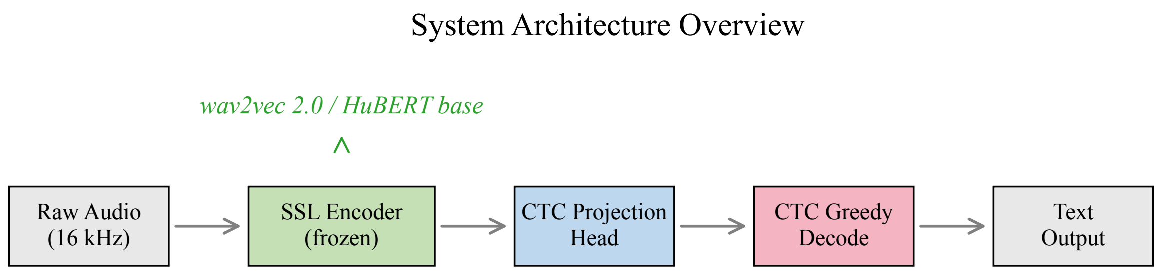

Fig. 1. System architecture. Raw audio passes through a frozen SSL encoder (wav2vec 2.0 or HuBERT base), a CTC projection head, and greedy CTC decoding.

materializes FLAC files locally, and writes JSONL manifests. The training split contains 269 utterances totaling 1.004 hours, sampled from train-clean-100. The development and test manifests contain 2703 and 2620 utterances from dev-clean and test-clean, respectively.

Transcripts are lowercased, punctuation is removed, apostrophes are preserved, and whitespace is collapsed. The CTC vocabulary is character-level with 30 tokens: blank, pad, unk, 26 letters a–z, apostrophe, and space. Vocabulary is built from the training split only; no external lexicon or language model is used.

2.3. Log-Mel CTC Baseline

The baseline extracts 80-dimensional log-mel filterbank features (25 ms Hamming window, 10 ms shift) via torchaudio, applies $\log(1+x)$ compression, and feeds the sequence to a feed-forward encoder with four hidden layers (256 units each, ReLU, dropout 0.1). A linear CTC head projects to character logits. This model has $\sim$0.33 M trainable parameters and contains no pretrained speech knowledge; it serves as the non-SSL control.

2.4. SSL CTC Models

The SSL systems use Hugging Face Transformers encoders with a compact CTC projection head. We evaluate two pretrained models: facebook/wav2vec2-base (95.6 M parameters) [?] and facebook/hubert-base-ls960 (95.0 M parameters) [?]. Both accept 16 kHz raw waveforms and output 768-dimensional hidden states with a 20 ms frame interval (320× CNN subsampling). The CTC projection head is Linear($768 \to 768$) $\to$ ReLU $\to$ Dropout(0.1) $\to$ Linear($768 \to V$), where V is the vocabulary size.

In all SSL runs, the encoder is frozen and only the projection head is trained ($\sim$1.2 M trainable parameters). This keeps the experiment feasible on an 8 GB GPU and isolates the value of pretrained representations.

For layer analysis, we compare wav2vec 2.0 final-layer (layer 12) features against intermediate-layer (layer 6) features, holding all other variables—dataset, optimizer, frozen encoder, CTC head, and decoding procedure—constant. This is a single-variable ablation.

2.5. Experimental Controls

All systems share the same text normalization, character vocabulary, CTC objective [?], and greedy CTC decoding. No external language model, beam search, or weight averaging is used. This makes the comparison conservative: an LM would improve absolute WER but would obscure how much information the acoustic representation itself contributes. The log-mel system is the non-pretrained control; the HuBERT and wav2vec 2.0 systems test the effect of SSL pretraining; and the wav2vec 2.0 layer-6 run isolates the hidden-layer choice.

3. EXPERIMENTAL SETUP

All training used PyTorch 2.7.1 with CUDA 12.8 on an NVIDIA GeForce RTX 4060 Laptop GPU (8 GB VRAM). Training hyperparameters are summarized in Table 1. All runs used the AdamW optimizer, gradient clipping at 1.0, and fixed random seed 1337 with no stochastic data augmentation.

WER and CER, computed via Levenshtein alignment at the word and character level, are the primary metrics. Real-time factor—inference wall time divided by audio duration—is a secondary efficiency metric. For SSL models, evaluation uses a 30-second maximum utterance duration, covering 2694/2703 dev and 2611/2620 test utterances. The log-mel run attains WER = CER = 1.000

Table 1. Training hyperparameters.

	Log-mel	wav2vec2 last	wav2vec2 layer 6	HuBERT last
Epochs	8	5	5	5
Batch size	8	2	2	2
Learning rate	3×10^{-4}	10^{-3}	10^{-3}	10^{-3}
Gradient clip	1.0	1.0	1.0	1.0
Max train audio	18 s	18 s	18 s	18 s
Max eval audio	30 s	30 s	30 s	30 s
Trainable params	0.33 M	1.2 M	1.2 M	1.2 M

under both 30-second and 60-second limits, so the exclusion does not affect any conclusion.

4. RESULTS AND ANALYSIS

4.1. Main Results

Table 2 presents the primary results. Fig. 2 shows training dynamics, and Fig. 3 provides a visual comparison of test-set error rates.

The log-mel baseline converges to empty or near-empty hypotheses for every utterance, yielding 1.000 WER and CER. This is the CTC blank-dominance failure mode: with one hour of supervision, the small feed-forward encoder cannot learn meaningful acoustic-phonetic mappings and collapses to predicting blank at every frame. The training loss decreases ($9.2 \rightarrow 2.9$) but reflects the model learning to output blank rather than speech content.

The final layer of frozen wav2vec 2.0 is somewhat better at the character level (CER 0.596) but word-level performance stays near the baseline (WER 0.985). Final-layer representations optimized for contrastive discrimination appear poorly aligned with the needs of a small character CTC head.

4.2. Layer Ablation

The most striking result is the layer ablation. As shown in Fig. 4, changing only the wav2vec 2.0 representation from the final layer to layer 6 improves test WER from 0.985 to 0.667 ($\Delta = -0.318$) and test CER from 0.596 to 0.262 ($\Delta = -0.334$)—with all other factors held constant.

This aligns with prior work [?] showing that intermediate SSL layers encode richer phonetic information, while final layers become specialized to the pretraining objective. Under frozen-encoder CTC training with limited data, the phonetic content preserved in middle layers is substantially more useful than the abstract final-layer representations.

4.3. Model Comparison

HuBERT base achieves the best overall performance (0.662 test WER, 0.236 test CER), narrowly ahead of wav2vec 2.0 layer 6 (0.667 test WER, 0.262 test CER). The difference is modest (ΔWER = 0.005, ΔCER = 0.026), suggesting that within the frozen-encoder low-resource regime, the choice of hidden layer can matter as much as the choice between SSL model families. HuBERT's advantage is consistent with its design: the masked-prediction objective with clustered acoustic units appears to produce representations that transfer more readily to phoneme-level tasks than wav2vec 2.0's contrastive objective.

Fig. 5 shows per-epoch development WER. HuBERT and wav2vec 2.0 layer 6 converge rapidly; by epoch 2 they reach WER below 0.75 and continue improving through epoch 5. The wav2vec 2.0 final-layer system plateaus at much higher WER, and the log-mel baseline never leaves the 1.000 WER region.

4.4. Inference Efficiency

All SSL systems achieve RTF in the range 4.1–4.4×10^{-3}: one second of audio is decoded in $\sim$4 ms on the RTX 4060, well below the real-time threshold. The log-mel baseline is even faster ($\sim 4\times10^{-5}$) due to its tiny model size, though the speed is irrelevant given 100% error.

4.5. Error Analysis

Fig. 6 shows the distribution of word-level error types. The log-mel baseline exhibits near-total deletion: 99% of reference words are absent from the hypothesis, confirming blank dominance.

Across SSL systems, substitutions account for the largest share (45–60%), followed by deletions (30–40%) and insertions (5–15%). The substitution-heavy profile is characteristic of character CTC decoding without a language model: some phonetic structure is captured, but character-level errors cascade into incorrect word hypotheses. Deletions are particularly common on short

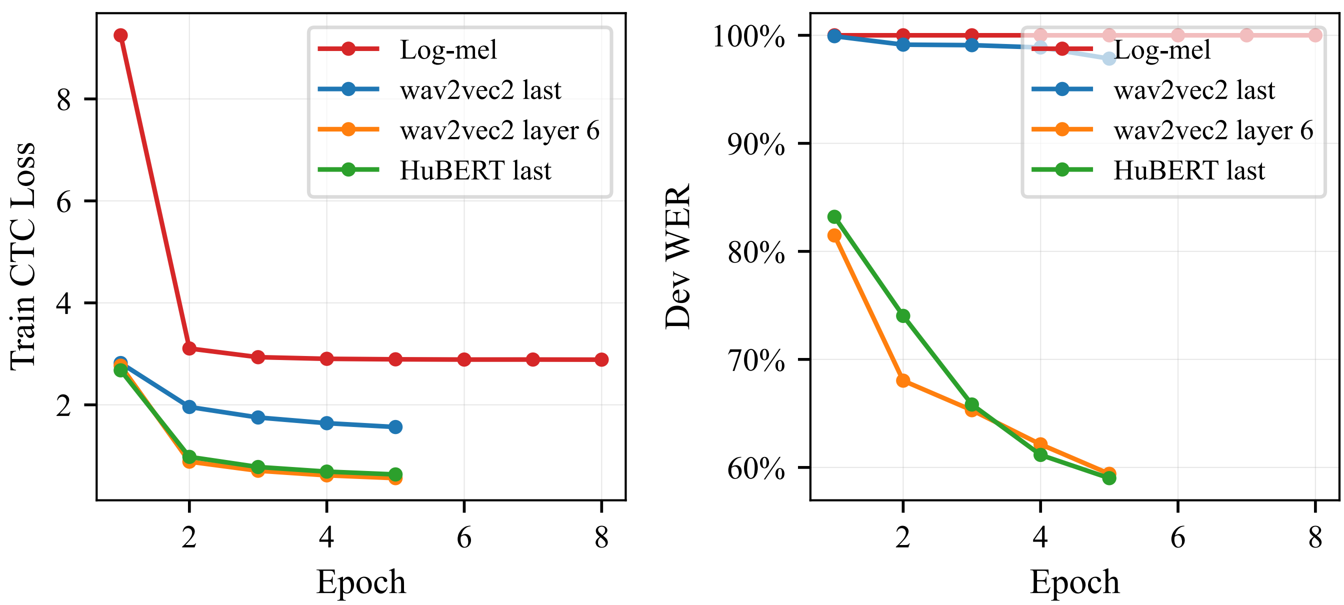

Fig. 2. Training dynamics. (Left) CTC training loss per epoch. (Right) Development WER per epoch. HuBERT and wav2vec 2.0 layer 6 converge rapidly and reach substantially lower dev WER.

Table 2. Low-resource ASR performance using 1 hour of LibriSpeech train-clean-100. All SSL encoders are frozen; only the CTC head is trained. RTF measured on an RTX 4060 Laptop GPU.

System	Dev WER ↓	Dev CER ↓	Test WER ↓	Test CER ↓	RTF ↓	#Params
Log-mel (baseline)	1.000	1.000	1.000	1.000	4.1×10^{-5}	0.33 M
wav2vec2 last	0.986	0.600	0.985	0.596	4.3×10^{-3}	95.6 M
wav2vec2 layer 6	0.659	0.255	0.667	0.262	4.2×10^{-3}	95.6 M
HuBERT last	0.656	0.232	0.662	0.236	4.4×10^{-3}	95.0 M

function words and rare content words.

Table 3 provides representative error examples. HuBERT errors often preserve phonetic structure—for instance, "stephanos dedalos" becomes "stdeffenos det lse"—but miss exact spelling. Short utterances and proper nouns are especially challenging without a language model.

5. DISCUSSION

5.1. Why Does Layer Choice Matter So Much?

The sharp performance gap between wav2vec 2.0 final and intermediate layers has a clear interpretation. Prior layer-wise analyses [?] show that intermediate SSL layers (around 6–9 in a 12-layer model) encode phonetic and articulatory features most explicitly, while final layers become increasingly invariant to surface acoustic detail and specialized toward the pretraining task. In wav2vec 2.0, the pretraining task is contrastive: the model learns to distinguish true quantized representations from distractors. Final representations optimized for this objective may discard precisely the fine-grained phonetic information a character CTC decoder needs. Intermediate layers retain richer acoustic-phonetic detail and thus provide a better starting point.

A practical implication: for practitioners using frozen SSL encoders in low-resource settings, exploring multiple hidden layers is a cheap hyperparameter search that can yield large gains without increased training cost.

5.2. Why Not Fine-Tune the Encoder?

Fine-tuning the SSL encoder would almost certainly improve results, but it was excluded from this study by design. The goal is to isolate and measure the value of frozen pretrained representations. Fine-tuning would conflate the value of pretraining with the value of supervised adaptation. Fine-tuning experiments are a natural next step.

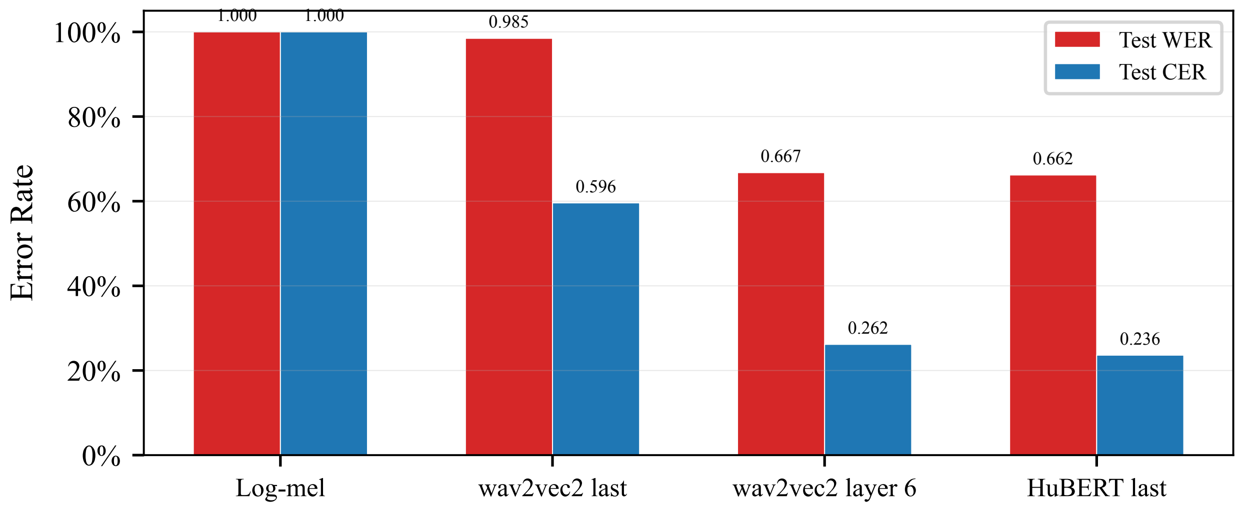

Fig. 3. Test-set WER and CER across all systems. The log-mel baseline collapses to empty hypotheses. HuBERT achieves the lowest error rates, followed closely by wav2vec 2.0 layer 6.

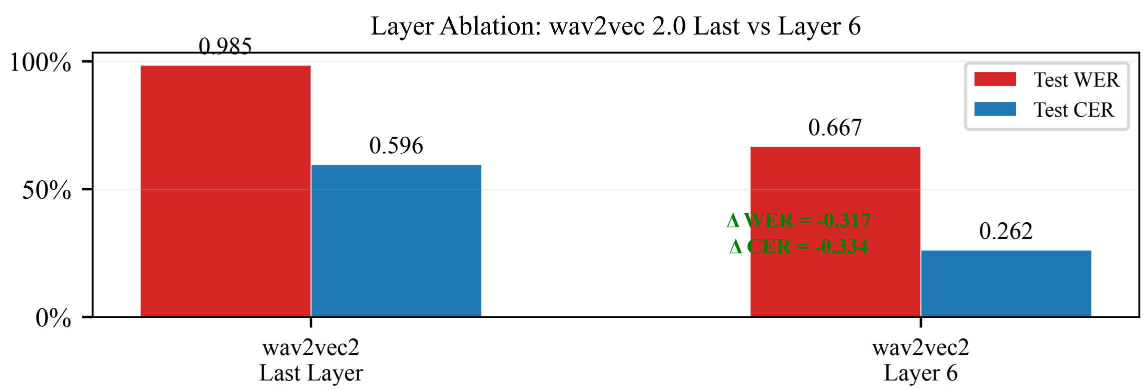

Fig. 4. Layer ablation. Switching from wav2vec 2.0 final layer to layer 6 reduces WER by 0.318 and CER by 0.334, holding all other variables fixed.

6. LIMITATIONS

This study is intentionally scoped to continuous SSL hidden states with frozen encoders. Several extensions are beyond the current scope:

- Discrete units: No vector quantization is applied, so token rate, bitrate, and codebook size are not reported.
- Encoder fine-tuning: All SSL experiments freeze the encoder. Fine-tuning with a small learning rate after a frozen warm-up is a standard recipe likely to improve results.
- Language model: No external LM, beam search, or shallow fusion is used. A 4-gram or neural LM would reduce WER, particularly for the substitution-heavy error profile observed in SSL systems.
- Model diversity: Only wav2vec 2.0 base and HuBERT base are compared. WavLM, XLS-R, or data2vec may exhibit different transfer properties.
- Data scale: Only the 1-hour point is evaluated. A systematic sweep over data quantities would reveal how the SSL vs. non-pretrained gap scales with labeled data availability.
- Robustness: Only clean LibriSpeech splits are evaluated. Performance on noisy, spontaneous, or accented speech remains untested.

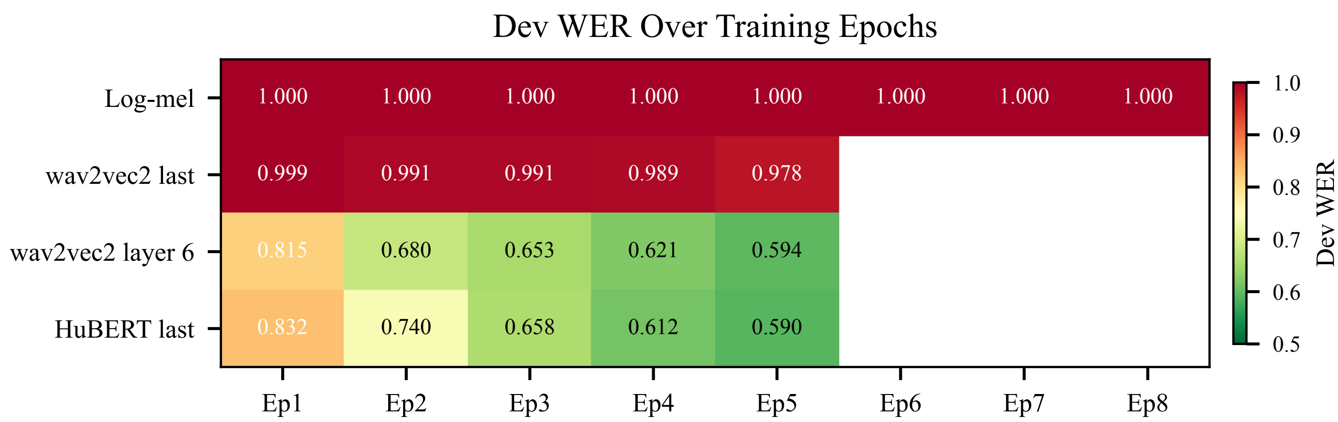

Fig. 5. Per-epoch development WER. Darker green = lower WER. HuBERT and wav2vec 2.0 layer 6 converge quickly, reaching WER below 0.6 by epoch 5.

Table 3. Representative error examples. HuBERT preserves phonetic structure but makes character errors. The log-mel baseline outputs empty strings.

Reference	HuBERT Prediction	Log-mel
squeak squeak	s cek s cqek	(empty)
stephanos dedalos	stdeffenos det lse	(empty)
farewell madam	ere wel madm	(empty)
he's swiftly punished	hese s wifly punihd	(empty)

7. REPRODUCIBILITY

The submission includes the complete source code (src/ and scripts/), YAML configurations (configs/), generated manifests (data/manifests/), per-system metrics and prediction CSVs (outputs/), error-analysis artifacts (results/), all figures (report/figures/), and the editable LaTeX source. Setup and run instructions are documented in README.md and REPRODUCIBILITY.md. All training is deterministic (fixed seed, frozen encoders, no data augmentation); re-running the provided scripts on the same hardware reproduces the reported numbers within floating-point precision.

8. CONCLUSION

This project implements a complete low-resource SSL ASR pipeline and evaluates four systems under a strict 1-hour labeled-data constraint. The key findings are:

1. SSL representations dramatically outperform a non-pretrained baseline. The log-mel baseline fails entirely (WER = 1.000), while HuBERT reaches 0.662 WER despite a frozen encoder.

2. Hidden layer choice can matter as much as model family choice. Switching wav2vec 2.0 from the final layer to layer 6 improves WER by 0.318 absolute, nearly matching HuBERT.

3. HuBERT base provides the best frozen representations (0.662 test WER, 0.236 test CER), though the margin over wav2vec 2.0 layer 6 is small.

4. Error analysis reveals substitution-dominated errors consistent with character CTC decoding without a language model. Phonetic structure is partially captured, but lexical modeling remains weak.

These results demonstrate that speech SSL pretraining is a powerful tool for low-resource ASR, and that representation selection—particularly the choice of hidden layer—is a critical design decision that can outweigh the choice of pretraining algorithm.

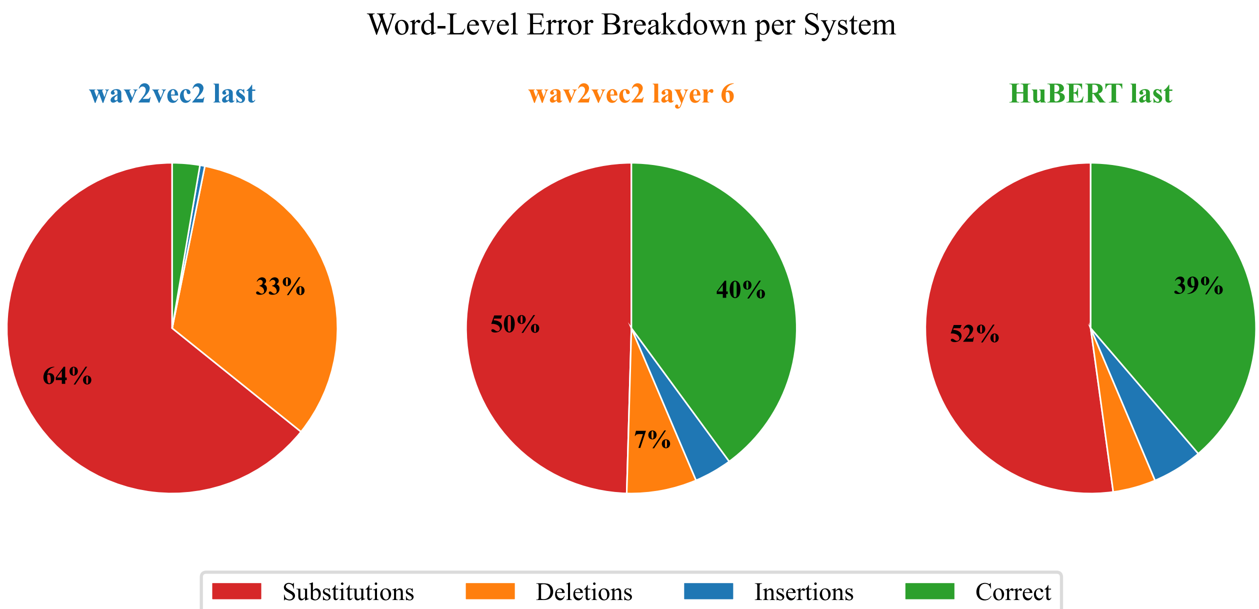

Fig. 6. Word-level error type distribution. Substitutions dominate across SSL systems (39–52%), followed by deletions (27–33%). Insertions are infrequent (8–16%). The log-mel system produces nearly all deletions.